Confidence-Shielded Evaluator-Conditioned Fusion for Risk-Controlled Infrared–Visible Object Detection

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Abstract

Infrared–visible image fusion is commonly treated as a preprocessing step whose output should benefit any downstream detector. In practice, the same fused image may improve one detector, leave another unchanged, and degrade a third because detection utility depends on the learned detector representation and confidence calibration. We introduce Confidence-Shielded Evaluator-Conditioned Fusion (CS-ECF), a label-free input-arbitration policy that evaluates visible, CDF, PIAFusion, and SPGFusion candidates with the target detector itself. CS-ECF averages each candidate's five highest detection confidences and permits a fused input only when its confidence exceeds that of the visible image by a factor of 1.5; otherwise, it falls back to the visible image. The margin was fixed on validation data before confirmatory testing. We evaluated the frozen policy on LLVIP, a grouped six-class M3FD test set, and three consecutive groups from the official DroneVehicle test split, comprising 12 detector checkpoints. Nine checkpoints improved, the dataset-balanced AP50 gain was 0.0133, and the model-by-group hierarchical 95% interval was [0.0028, 0.0213]. Removing the shield introduced a worst AP50 loss of 0.0584, whereas the frozen policy limited the worst loss to 0.0040. The two DroneVehicle checkpoints improved by 0.0251 and 0.0293, although they belong to the same training run. M3FD gains were concentrated in the people class, and LLVIP intervals crossed zero. These results support CS-ECF as a risk-controlled evaluator-aware fusion-input system for pedestrian-centric and mixed infrared–visible detection; they do not support

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uniform improvement across all object classes. Under a standardized serial benchmark, candidate generation and four detector evaluations required approximately 0.71 s per image pair, dominated by SPGFusion.

Keywords: infrared–visible image fusion, object detection, evaluator-conditioned fusion, confidence calibration, risk control, thermal imaging

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1. Introduction

Infrared–visible imaging combines the spatial detail and color of visible sensors with thermal evidence that can remain available when illumination is poor, smoke reduces contrast, or targets have weak visual texture. This complementarity has motivated pixel-, feature-, and network-based fusion methods for surveillance, autonomous perception, and thermal imaging systems. [1–4] In detection-oriented applications, the value of a fused image is not determined by visual appeal alone; it is the change in detector outputs under a specified operating condition.

Current practice often selects one fused representation and evaluates it with one detector on one benchmark. This design assumes that a globally preferred fusion result remains preferable across detector architectures and training seeds. The assumption is fragile because detectors differ in receptive fields, confidence calibration, training data, and sensitivity to high-frequency detail. Consequently, a fusion rule that improves one evaluator may reduce the average precision (AP) of another on the same image.

Per-image selection offers a natural response: run several fusion candidates and choose the one with the strongest detector response. Yet unconstrained selection converts small calibration differences into frequent input switches and can produce severe losses when a spurious high-confidence response is mistaken for improved detection. The problem is therefore not only to predict which fused image looks best, but to decide when the evidence is strong enough to depart from the visible-image safety baseline.

Recent infrared–visible fusion and detection studies have optimized shared fusion networks, cross-modal feature fusion, or hierarchical detectors [5–8]. These studies establish the relevance of joint fusion and detection, but they generally train or optimize a detector-fusion system. They do not establish a label-free arbitration rule whose switching threshold is fixed independently of the confirmatory tests and whose behavior is reported at the level of detector checkpoint, image, and scene or frame group.

We formulate infrared–visible fusion selection as evaluator-conditioned input arbitration. CS-ECF evaluates four candidates with the target detector, computes a top-five confidence score, and accepts a fused candidate only when its score reaches 1.5 times that of the visible input. The threshold is selected on validation data and frozen before testing. We evaluate the policy on LLVIP, M3FD, and DroneVehicle with model-, image-, and group-aware uncertainty, compare it with fixed fusion and an unshielded selector, and report the cost and failure boundaries. The contributions are:

1. an evaluator-conditioned formulation of fusion-input selection that does not use test labels;
2. a confidence shield that reduces catastrophic switching while retaining positive dataset-balanced gains;
3. a three-dataset confirmation that separates image-level uncertainty, scene/group uncertainty, and checkpoint variability, including explicit evidence that the M3FD gain is concentrated in the people class.

2. Methods

2.1. Problem formulation

Let x_v and x_i denote a registered visible–infrared pair, and let $\mathcal{D}$ denote the target detector checkpoint. A candidate $c \in \mathcal{C}$ produces an input image x_c, where $\mathcal{C} = \{\text{vis}, \text{cdf}, \text{pia}, \text{spg}\}$. The policy returns one candidate per pair without access to ground-truth boxes:

$$g(x_v, x_i; \mathcal{D}) \in \mathcal{C}. \tag{1}$$

The final detection output is $\mathcal{D}(x_{g(x_v,x_i;\mathcal{D})})$. The detector is therefore reused for candidate evaluation and final inference; it is not an external label oracle.

2.2. Fusion candidates

The visible input is always included as a safety candidate. CDF combines a pixel-mean image and an IFCNN image through sign-consistent high-frequency details:

$$d(x) = \begin{cases} \operatorname{sign}(d_m + d_i)\min(|d_m|, |d_i|), & d_m d_i > 0, \\ 0, & \text{otherwise.} \end{cases} \tag{2}$$

A thermal-validity mask M is formed from the infrared top-hat response, infrared intensity, visible luminance, connected components, and smoothing. The shared detail is injected into the LAB luminance channel:

$$L' = L + \alpha M d. \tag{3}$$

Table 1: Datasets and confirmatory evaluation units.

Dataset	Split	Images	Classes	Evaluation groups
LLVIP	Official test	300	1	7 scenes
M3FD	Grouped test	200	6	2 frame groups
DroneVehicle	Official test subset	300	1 OBB vehicle	3 frame groups

The frozen settings are $q = 0.95$, $\alpha = 0.20$, and $\sigma = 5$. PIAFusion and SPGFusion are generated with their official model definitions and pretrained weights under the same image-pair protocol. [2–4]

2.3. Confidence-shielded arbitration

For candidate c, let $s_{c,1} \geq s_{c,2} \geq \cdots$ be detector confidences. We define

$$q_c = \frac{1}{\min(K, n_c)} \sum_{j=1}^{\min(K,n_c)} s_{c,j}, \qquad K = 5, \tag{4}$$

where n_c is the number of detections. Candidates with no detections receive $q_c = -1$. The highest-scoring candidate is $c^* = \arg\max_c q_c$, with ties resolved in favor of visible. The selected input is

$$c = \begin{cases} c^*, & c^* \neq \text{vis and } q_{c^*} \geq 1.5 q_{\text{vis}}, \\ \text{vis}, & \text{otherwise.} \end{cases} \tag{5}$$

The margin 1.5 was selected on validation data before the confirmatory tests. The selector uses no test labels and makes no class-specific decision.

2.4. Datasets and evaluators

LLVIP supplies a 300-image pedestrian test set spanning seven scenes [9]. M3FD supplies a grouped six-class test set of 200 images [10], split by consecutive frame groups to reduce adjacent-frame leakage; the detector families are YOLO26n and YOLOv8n. DroneVehicle contributes 300 image pairs from three fixed consecutive 100-frame groups of the official test archive. The groups were selected with a predefined seed of 2037 before evaluation. The YOLO11x-OBB best and last checkpoints were both evaluated. The source archive has 4,430,343,899 bytes and SHA-256 `94BF47E493E7A57FD5D900439F614F339E7B6C7181E4D781BCFEBAEB963FFD8E` [11]. All OBB outputs use a confidence floor of 0.001 and an NMS IoU threshold of 0.7.

2.5. Statistical analysis

Point estimates use pooled AP50 after concatenating detections across test images within a checkpoint. Image bootstrap intervals resample images within a checkpoint. Group bootstrap intervals resample scenes or consecutive frame groups and are the primary clustered intervals. The dataset-balanced estimand assigns equal weight to LLVIP, M3FD, and DroneVehicle and resamples checkpoints and groups. M3FD has only two test groups and LLVIP has seven scenes, so the group bootstrap is reported as a finite-cluster analysis. The model-by-group bootstrap used 500 resamples with `numpy.random.default_rng(2219)` in Python 3.10.22 and NumPy 2.2.6. Spearman association tests used SciPy 1.15.3.

2.6. Implementation and cost

At a standardized fusion input of 384×512, FP32 and batch size one on an NVIDIA RTX 2080 Ti, the CDF pipeline required 22.3 ms, PIAFusion 37.7 ms, and SPGFusion 516.3 ms. Four serial YOLO11x-OBB evaluations at an input size of 640 required 130.0 ms, giving an estimated serial total of 706.3 ms. This estimate excludes model loading or swapping, data loading, and final detection post-processing. SPGFusion's additional peak activation memory was 1453.5 MiB; per-stage memory values must not be summed as an end-to-end peak.

3. Results

3.1. Fusion utility is evaluator-dependent

Across the 12 confirmatory checkpoints, the same candidate library produced both positive and negative gains (Fig. 1). CS-ECF improved nine checkpoints. Its dataset-balanced AP50 gain was 0.0133, with a model-by-group hierarchical 95% interval of [0.0028, 0.0213]. The unshielded selector improved seven of 12 checkpoints, had a larger dataset-balanced mean of 0.0332, and produced a worst loss of 0.0584.

3.2. Dataset-specific behavior

On LLVIP, the mean gain was 0.0016 and five of eight checkpoints improved; the model and image intervals crossed zero. On M3FD, both six-class checkpoints improved, with a mean gain of 0.0111. On the independent DroneVehicle test subset, the best and last OBB checkpoints improved by 0.0251 and 0.0293; the combined group interval was [0.0002, 0.0406]. All three DroneVehicle groups were non-negative, but one gained only 0.0002. The two OBB checkpoints originated from the same training run and are not independent architectures.

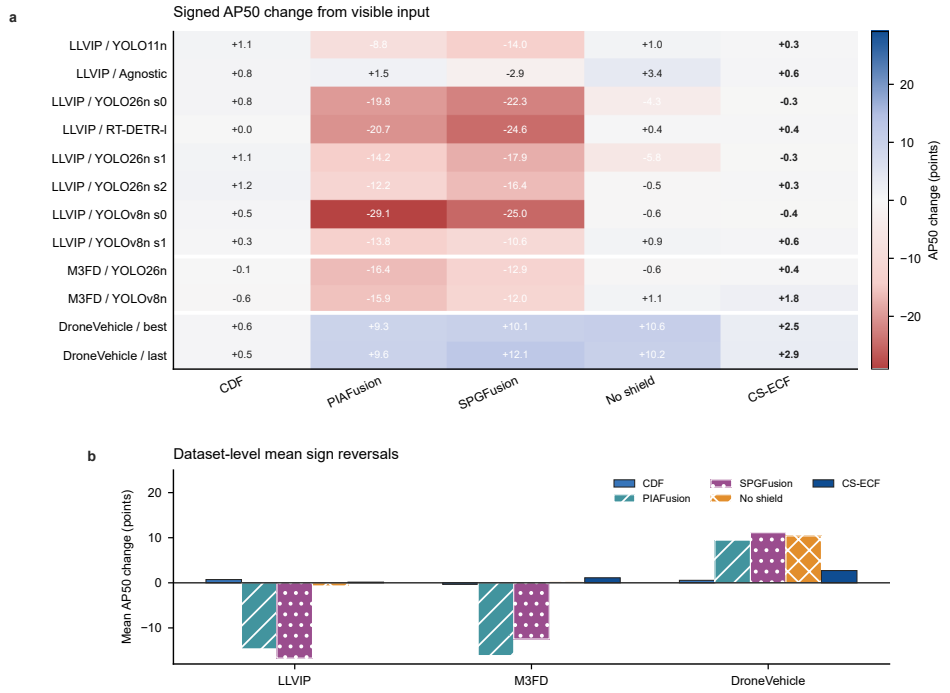


Figure 1: **Evaluator-dependent utility of infrared–visible fusion candidates. a**, Signed AP50 change from the visible input for CDF, PIAFusion, SPGFusion, the unshielded selector, and CS-ECF across 12 detector checkpoints. Cells are point contrasts. **b**, Dataset-level mean AP50 changes. Equal candidate sets produce opposite signs across datasets. Source data are provided in `figure1_source.csv`.

3.3. The shield is a risk-control mechanism

Without the shield, the policy selected fused candidates on most LLVIP and DroneVehicle images and occasionally produced large confidence gains that did not translate into AP. The 1.5 margin reduced switches and lowered the worst AP50 loss from 0.0584 to 0.0040. On DroneVehicle, however, the shield also reduced the mean gain from 0.1040 to 0.0272. This trade-off is the central quantitative meaning of risk control in this study.

3.4. Fixed baselines and class concentration

Always CDF improved 10 of 12 checkpoints and had a dataset-balanced gain of 0.0031, but remained negative on both M3FD detectors. Always PIAFusion and Always SPGFusion had negative dataset-balanced gains because their strong DroneVehicle performance did not transfer to the other

Table 2: Confirmatory checkpoint summary and policy comparison.

Policy	Positive	Mean	Dataset-balanced	Worst
Always CDF	10/12	+0.0052	+0.0031	-0.0059
Always PIAFusion	3/12	-0.1087	-0.0710	-0.2910
Always SPGFusion	2/12	-0.1136	-0.0602	-0.2501
Top-5 no shield	7/12	+0.0132	+0.0332	-0.0584
CS-ECF	9/12	+0.0074	+0.0133	-0.0040

datasets. In the M3FD six-class evaluation, the CS-ECF gain was concentrated in people: YOLO26n improved by 0.0214 and YOLOv8n by 0.1090, while the other five classes changed by at most 0.0003 in absolute AP50 (Fig. 4). The evidence therefore supports a pedestrian/people-focused interpretation, not uniform six-class improvement.

3.5. Thermal feature association and mask ablation

On LLVIP, infrared top-hat contrast showed a weak positive association with switching (Spearman $\rho = 0.151$, $p = 0.009$), whereas visible luminance was only weakly negative ($\rho = -0.104$, $p = 0.071$). On M3FD, thermal contrast showed a moderate positive association ($\rho = 0.320$, $p < 10^{-5}$). On DroneVehicle, however, both thermal contrast ($\rho = -0.403$) and visible luminance ($\rho = -0.441$) were negatively associated with switching. A high-thermal/low-visible median split did not produce a consistently larger AP gain than the remaining images.

A fixed-input mask ablation gave a smaller but consistent thermal effect on DroneVehicle. Replacing the thermal-validity mask with global detail injection changed the CDF gain from 0.0060 to 0.0057 for the best checkpoint and from 0.0050 to 0.0038 for the last checkpoint. The thermal mask added 0.0007 AP50 on average relative to global injection.

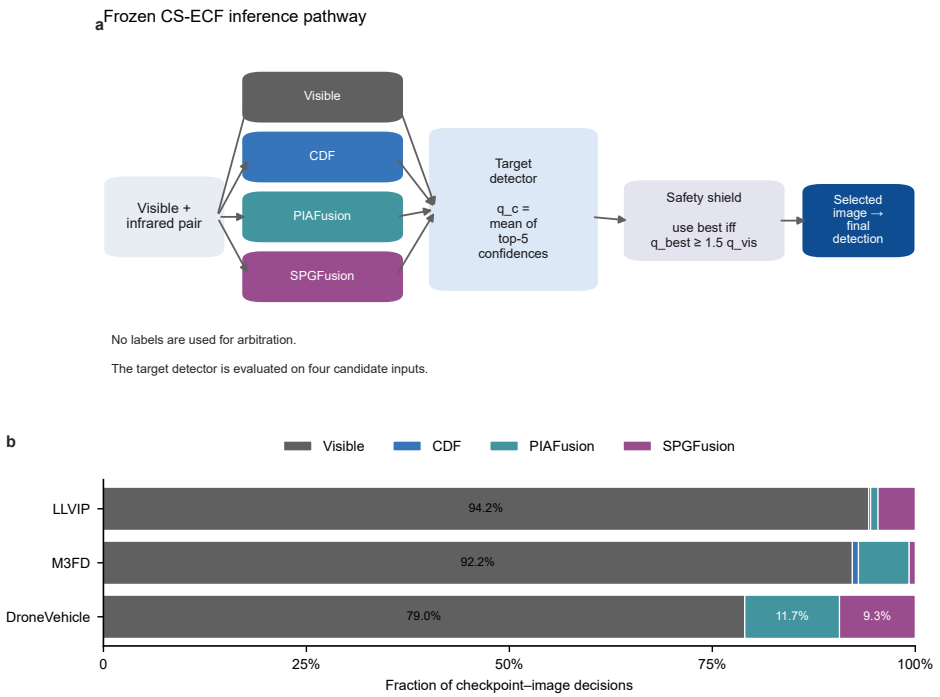


Figure 2: **Confidence-shielded evaluator-conditioned fusion.** **a**, The visible–infrared pair is passed to visible, CDF, PIAFusion, and SPGFusion candidates. The target detector computes its top-five mean confidence for each candidate. A fused candidate is selected only when its score is at least 1.5 times the visible score; otherwise the visible input is retained. No labels are used for arbitration. **b**, Composition of the frozen decisions across all checkpoint–image trials. Source data are provided in `figure2_source.csv`.

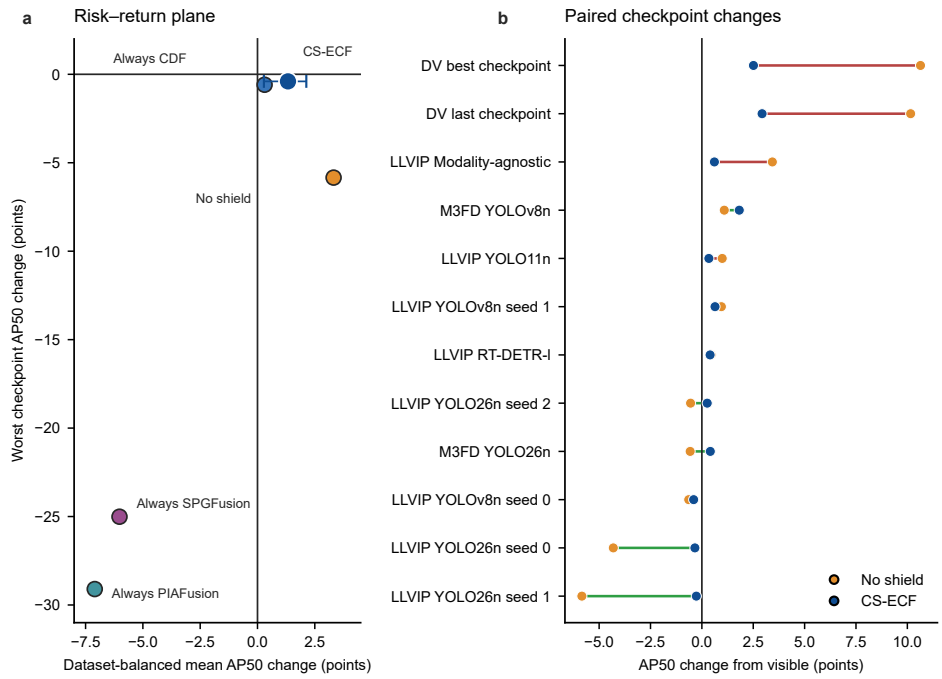


Figure 3: **Risk–return trade-off of fusion-input policies. a**, Dataset-balanced mean AP50 change versus worst-checkpoint AP50 change. The CS-ECF horizontal error bar is the model-by-group hierarchical 95% interval. **b**, Paired checkpoint changes under the unshielded selector and CS-ECF. The shield sacrifices part of the DroneVehicle mean gain to reduce the worst LLVIP loss. Source data are provided in `figure3_source.csv`.

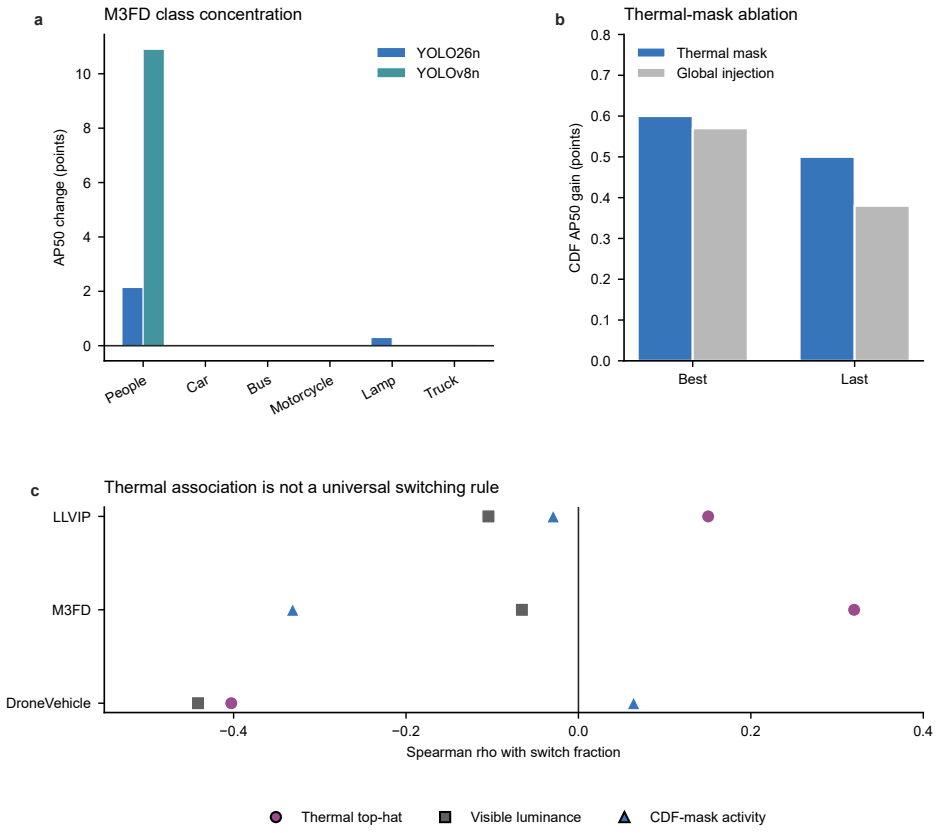


Figure 4: **Thermal constraints and class-specific effects.** **a**, M3FD AP50 changes for six classes. The gain is concentrated in people. **b**, Fixed-input DroneVehicle CDF gain with the thermal-validity mask or global detail injection. **c**, Spearman association between the fused-selection fraction and thermal top-hat contrast, visible luminance, or CDF-mask activity. The association reverses on DroneVehicle, which does not support a universal thermal switching rule. Source data are provided in `figure4_source.csv`.

4. Discussion

The central finding is that fusion utility is not an intrinsic property of a fused image alone. A candidate can improve one detector's confidence ordering while harming another's, and a policy that ignores this dependence can select inputs that appear confident but do not improve AP. CS-ECF responds by using the target detector as an input arbiter and requiring a large visible-relative margin before switching.

The shield's value is clearest in the tail. The unshielded selector achieved a higher dataset-balanced average on DroneVehicle, where both PIAFusion and SPGFusion were strong, but it also produced severe LLVIP losses. The 1.5 margin deliberately favors visible fallback under ambiguous evidence. This is consistent with a deployment rule in which a small probability of a large regression is more costly than a missed opportunity for a moderate gain.

Two boundaries constrain the interpretation. First, the strongest M3FD effect is class-specific: people detection improves substantially while the remaining classes are nearly unchanged. The selected thermal evidence is therefore most useful for a subset of target appearances, not all six vehicle classes. Second, the DroneVehicle confirmation uses two checkpoints from the same training run. The official test groups are independent of model selection, but the checkpoints share training data and are correlated.

The thermal stratification adds another limitation. Thermal contrast may increase the likelihood of selecting fused evidence in LLVIP and M3FD, but the association reverses on DroneVehicle. Sensor-level thermal evidence and detector confidence therefore capture different information. Neither should be interpreted as a universal utility score. The appropriate claim is thermal fusion utility under a detector-conditioned decision rule, not thermal dominance.

The cost analysis also matters. SPGFusion dominates latency, and four detector evaluations are not free. CS-ECF is best viewed as a risk-controlled system for settings in which the target detector is already run and the cost of a catastrophic input choice is high. It is not a low-latency replacement for a single fusion operator.

5. Conclusions

We showed that infrared–visible fusion utility is evaluator-conditioned and proposed CS-ECF, a confidence-shielded input-arbitration policy that falls back to visible unless a detector-supported fusion candidate clears a

fixed visible-relative margin. Across three test datasets and 12 checkpoints, the policy improved nine checkpoints, produced a dataset-balanced AP50 gain of 0.0133, and reduced the worst observed degradation from 0.0584 to 0.0040. The evidence supports risk-controlled, pedestrian-centric infrared–visible detection; it does not support uniform six-class improvement, and the additional cost of SPGFusion and four detector evaluations must be considered in deployment.

CRediT authorship contribution statement

Jianyong Zhao: Conceptualization, Methodology, Software, Formal analysis, Writing – original draft. **Zigao Wu:** Supervision, Writing – review & editing. **Mengfei Ma:** Investigation, Validation, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this article.

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Data availability

The underlying datasets are publicly available. LLVIP is distributed at `https://github.com/bupt-ai-cz/LLVIP`. M3FD is distributed with the TarDAL benchmark at `https://github.com/JinyuanLiu-CV/TarDAL`. DroneVehicle is distributed through `https://github.com/VisDrone/DroneVehicle` and was obtained from the public archive `https://huggingface.co/datasets/McCheng/DroneVehicle`. Each dataset is used under the licence stated by its distributor. The fixed DroneVehicle test-group manifest, source-archive size and SHA-256, CS-ECF per-image score files, source-data tables, and scripts that regenerate the reported intervals are included in the public analysis repository described below.

Code availability

The code for reproducing the experiments is available at `https://github.com/lessThanbetter/ssgf` and is archived at `https://doi.org/10.5281/zenodo.22984389` under an MIT licence. The repository contains the fusion and arbitration code, detector-evaluation scripts, fixed split and test manifests, per-image score files, result tables, figure source data, and the numerical checks used to generate the reported intervals. The versioned archive should be updated from the submission tag before publication so that the DOI resolves to the same release used for the manuscript.

Use of generative artificial intelligence

During the preparation of this work the authors used AI-assisted tools to support language editing, data-analysis code development, figure preparation, and the organization of numerical results. The authors reviewed and edited the generated material, verified the reported analyses and figures against the source data, and take full responsibility for the content of the publication.

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